

Math 520, F07
Quiz 5

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. You must simplify your answer if possible.

Problem 1. (10 points) Solve the following initial-value problem

$$2y'' - 3y' - 5y = 0; \quad y(1) = 3, \quad y'(1) = 4.$$

Solution: Characteristic equation is

$$2r^2 - 3r - 5 = 0$$

$$(r+1)(2r-5) = 0$$

$$r_1 = -1, \quad r_2 = \frac{5}{2}$$

Hence, we have

$$y(t) = C_1 e^{-(t-1)} + C_2 e^{\frac{5}{2}(t-1)} \quad (\text{Since } t_0 = 1)$$

is the general solution. From the initial conditions, we know

$$y(1) = 3 \Rightarrow \underline{C_1 + C_2 = 3} \quad - \textcircled{1}$$

$$y'(t) = -C_1 e^{-(t-1)} + \frac{5}{2} C_2 e^{\frac{5}{2}(t-1)}$$

$$y'(1) = 4 \Rightarrow \underline{-C_1 + \frac{5}{2} C_2 = 4} \quad - \textcircled{2}$$

Then $\textcircled{1}$ & $\textcircled{2}$ gave us $C_1 = 1, C_2 = 2$

Thus,

$$y(t) = e^{-(t-1)} + 2e^{\frac{5}{2}(t-1)}.$$

Problem 2. (10 points) Solve the following initial-value problem

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = 0; \quad y(0) = 0, \quad y'(0) = 2.$$

Solution: Characteristic equation is

$$r^2 + 2r + 5 = 0$$

$$r_1 = -1 + 2i, \quad r_2 = -1 - 2i.$$

So that we know

$$y_1(t) = e^{-t} \cos 2t \text{ and } y_2(t) = e^{-t} \sin 2t$$

are two independent solutions of this equation.

Then,

$$\begin{aligned} y(t) &= C_1 y_1(t) + C_2 y_2(t) \\ &= e^{-t} [C_1 \cos 2t + C_2 \sin 2t] \end{aligned}$$

is the general solution. From the initial conditions,

$$y(0) = 0 \Rightarrow \underline{C_1 = 0} \quad - \textcircled{1}$$

$$y'(t) = -e^{-t} [C_1 \cos 2t + C_2 \sin 2t] + e^{-t} [-2C_1 \sin 2t + 2C_2 \cos 2t]$$

$$y'(0) = 2 \Rightarrow \underline{-C_1 + 2C_2 = 2} \quad - \textcircled{2}$$

Then, $\textcircled{1}$ & $\textcircled{2}$ give us $C_1 = 0, C_2 = 1$

Thus,

$$y(t) = e^{-t} \sin 2t$$